

L1.1 — Why Local AI?

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Level: Foundation (Free) · **Time:** 20 minutes · **Type:** Theory — no installation, no terminal
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[Objectives] What you'll know by the end

- The real difference between cloud AI and local AI
 - 3 concrete reasons why local AI is the right choice
 - How LeYa Academy's architecture actually works — what's local and what's API
 - Exactly what you'll have built after these first 5 lessons
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[Content] Content

1. Where is your AI right now?

In 2026, when you use ChatGPT, Claude, or Gemini, here's what happens:

- You type a question
- Your text travels over the internet to a server in Virginia, Frankfurt, or Singapore
- A massive AI model processes your question
- The answer comes back to you

Sounds simple. But:

- **You pay monthly.** \$20/month for ChatGPT Plus. \$240/year. Forever.
- **Your data sits on their servers.** Conversations, projects, ideas — all stored and analyzed.
- **If their server goes down, your AI disappears.** Nothing you can do. You wait.
- **You have no control.** Can't change how it thinks, what it remembers, how it responds.

The local model flips all of this.

2. Cloud vs Local — the real comparison

	Cloud AI (ChatGPT, Claude)	Local AI (what you're building)
Cost	\$20/month × 12 = \$240/year	\$0 — free software, only the API costs (~\$0.50/million tokens)
Your data	On their servers	On your laptop
Internet	Required	Optional — Hermes works offline; only the API needs internet
Control	Zero	Total — you change anything
Availability	When their servers are up	24/7, as long as your laptop is on
Privacy	Your conversations are training data	No one sees them, no one accesses them

Why does privacy matter? Because anything you type into a cloud chatbot can be used to train the next version of the model. Your private conversation from yesterday = part of tomorrow’s training data. On a local AI, your data never leaves your laptop.

3. What you’re building — the final product after 5 lessons

By the end of Level 1, you’ll have:

- **An AI assistant with its own personality** — you decide how it speaks, what it knows, what it refuses
- **Running on your laptop** — no browser, no cloud, from the terminal
- **That remembers who you are** — your name, your projects, your preferences
- **That no one else can access** — it’s local, it’s encrypted, it’s yours

And most importantly: **you built every component yourself**. You didn’t pay \$20 and get a chatbot. You installed Linux, configured Hermes, connected an API, wrote your first system prompt. By the end, you know what’s under the hood.

(!) **Think you need a supercomputer? You don’t.** Any laptop from the last 5-7 years works. If your laptop boots and has a USB port, you can do this course. Full hardware details in Lesson 2.

4. [Architecture] How It Really Works — LeYa Academy Architecture

You might be wondering: “if everything is local, why do I need the DeepSeek API in Lesson 5?” Fair question. Let’s clear it up.

Hermes Agent is the hub, not the brain.

Think of Hermes as an operating system for AI. It runs on your laptop — 100% local. But like any operating system, it can run applications that connect to the internet when needed.

Component	Where it runs	What it does
Hermes Agent	[Local] Local (your laptop)	Orchestrates everything: memory, skills, conversations, cron jobs
Your data	[Local] Local (your SSD)	Conversations, preferences, memory — nothing leaves
The AI model	[API] API (DeepSeek) OR [Local] Local	Processes questions and generates answers

Why start with DeepSeek API (cloud) instead of a local model?

1. **Simple.** No need to download a 2.4 GB model. No compiling llama.cpp. Just paste an API key.
2. **Fast.** Local models run slower on regular hardware. DeepSeek responds in 1-2 seconds.
3. **Capable.** DeepSeek V4 is one of the best models available. Small local models (2-3B parameters) are weaker.
4. **Cheap.** ~\$0.50 per MILLION tokens — that’s roughly 750,000 words processed. With \$5 in starting credit, you’ll have conversations for months.

What’s local and what’s API? — Honest breakdown:

Is it local?	Component
[x] Yes	Hermes Agent (the application)
[x] Yes	Your memory (conversations, preferences)
[x] Yes	Configuration files
[x] Yes	Skills, cron jobs, watchdog
[API] API	DeepSeek (processes responses — Level 1 & 2)
[x] Yes	Local model Gemma/Llama (processes responses — Level 3)

The long-term plan:

Level 1: Hermes (local) + DeepSeek API (cloud) → fast, functional AI
Level 2: Hermes (local) + DeepSeek API → advanced customization
Level 3: Hermes (local) + LOCAL model → full autonomy, zero cloud
Level 4: ESP32, voice, multi-agent → complete 100% local ecosystem

In short: You start with the API to get going fast. Your data stays on your laptop — only the question text and the answer pass through DeepSeek (encrypted, via HTTPS). By Level 3, you can replace the API with a local model, and then EVERYTHING is on your machine. Zero data to the outside.

If this sounds like an acceptable trade-off — keep going. If you want 100% local from day one — it's possible, but we recommend doing Levels 1-2 with the API and transitioning to local at Level 3.

5. What you need

You need	Details	Cost
Laptop	Any model from the last 5-7 years, 8GB RAM+, USB port	\$0 (you already have one)
External SSD	256GB+, recommended: Samsung T7 / SanDisk Extreme	~\$25-50
USB stick	8GB+ for Linux installation	~\$5
Time	~2 hours of focused attention	—
Internet	Only for initial downloads + API	—

That's it. You don't need to know Linux. You don't need to know programming. These lessons are written for zero experience.

[Exercise] Exercise

Calculate what you'd pay per year for the cloud version vs the local one:

ChatGPT Plus: \$20 × 12 months = \$240/year (and it goes up every year)
Local AI: \$0 (software) + ~\$50 (SSD, one-time) + ~\$12/year (API for normal use)
First year: cloud = \$240 | local = \$62
Difference: \$178 in your pocket - in the first year alone
 \$240 every year after that (you already have the SSD)

Write down the difference. It's real.

[Checkpoint] Checkpoint — don't move on until you can answer

1. What are the 3 main differences between cloud AI and local AI?

Expected: Cost (monthly subscription vs free + cheap API), privacy (their data vs your data), control (zero vs total).

2. Why is your data safer with local AI?

Expected: Because it never leaves your laptop. It's not stored on anyone else's servers, not used for training, not accessible to anyone. Only the current question text passes through the API (encrypted) — memory, preferences, and history stay local.

3. What will you have built by the end of these 5 lessons?

Expected: A working AI assistant, with personality, running locally via Hermes Agent, that remembers who you are, and that only you can access.

4. (bonus) Why start with DeepSeek API instead of a local model right away?

Expected: Because it's simpler, faster, more capable, and cheaper as a first step. The local model comes at Level 3.

(!!) **Don't move on until you can answer at least 3 out of 4.** If you didn't get them right: re-read sections 2, 3, and 4. This is the foundation for everything that follows. If it's still unclear after a second read — reach out to support.

All answers clear? Move on to L1.2 — Hardware: What You Need.

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